AEISU: Tensor Core-Accelerated Inverse Supersampling for Real-Time Game Upscaling

Charles Dale B. Gleason
Independent Researcher

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Abstract. We present AEISU, a compact $2\times$ spatial upscaler that predicts adaptive filters below the input grid and reconstructs directly from source RGB. A fused mixed-FP8/FP16 implementation achieves 0.132 ms median resident latency for 540p-to-1080p and 0.416 ms for 1080p-to-4K on RTX 5090. Evaluation on 600 QRISP pairs across five environments tests area, bicubic, and independently rendered inputs with and without spatial antialiasing. The area-trained model leads Area PSNR/SSIM and has lower bright overshoot and ringing than six neural comparators on target-selected isolated edges. Four neural comparators obtain lower mean Area LPIPS and better bicubic reconstruction; conservative spatial methods have lower edge excursions. On Native, PSNR falls below bilinear and EASU despite the lowest mean LPIPS, whose lead over the closest methods is unresolved. A separately DIV2K-trained L1 checkpoint exceeds RepNet and RT4KSR in mean fidelity and LPIPS on four standard bicubic benchmarks, while higher-compute neural models remain ahead. On the shared backbone, the adaptive head achieves higher FP16 fidelity and smaller FP8 degradation than the tested RGB-residual and depth-to-space heads.

1 Introduction

Single-image super-resolution (SISR) estimates a high-resolution (HR) image from a low-resolution (LR) observation. Because downsampling discards information, reconstruction depends on both image priors and the observation model. Convolutional, residual, and transformer methods learn this mapping from paired images [1, 3, 4].

We focus on *inverse supersampling* for resolved game frames. A common 8K master is independently area-resolved to a 1080p observation and a 4K target. The objective is to reconstruct the 4K resolve from the 1080p observation. This sampling model differs from bicubic image reduction and from rendering a scene independently at two resolutions.

At 1080p input, even narrow full-LR feature maps incur substantial arithmetic and memory traffic. AEISU keeps convolutional features below the LR grid and uses them to predict adaptive filters applied to the original RGB samples. Its channel widths, layouts, and precision schedule are designed for Tensor Core convolution; the local RGB reconstruction uses conventional CUDA arithmetic. We measure the complete resident image-to-image operation, since operation counts alone do not determine latency [5, 6, 17].

Our contributions are:

1. source-conditioned polyphase reconstruction with convolutional feature processing below the input grid;
2. folded mixed-precision execution, with output-head, core-architecture, and training-degradation controls;
3. multi-condition game evaluation characterizing degradation sensitivity, distortion, and LPIPS perceptual distance.

2 Related Work

Learned and efficient image super-resolution. SRCNN, ESPCN, EDSR, and SwinIR represent convolutional mapping, LR-grid subpixel reconstruction, residual learning, and transformer restoration, respectively [1, 2, 3, 4]. Their bicubic evaluations use a different observation model from our supersample-resolve task. Efficient SR methods also emphasize measured execution: SPAN reports quality, latency, and model complexity, while NTIRE evaluates these quantities under common protocols [5, 6]. RepNet uses a reparameterized convolutional design for real-time $4\times$ reconstruction [19]. RT4KSR reduces deep-feature resolution using pixel unshuffle and structural reparameterization [20]. QuickSRNet targets real-time mobile upscaling and robustness to quantization [8].

Adaptive reconstruction. Learned filtering is established in RAISR [7]. LAPAR predicts coefficients over predefined Gaussian/DoG bases and applies assembled filters to a bicubic-upsampled image [12]. Deep Adaptive Image Resampling predicts spatially varying kernels, and DUF predicts upsampling filters from video neighborhoods together with a separate residual [13, 14]. Kernel Prediction Networks also reconstruct directly from source samples, in their case for burst denoising [9]. HDRNet and deep guided filtering couple lower-resolution prediction to source-guided image processing [10, 11]. AEISU uses a single image, a learned four-phase base, sixteen normalized zero-DC components, and shared RGB weights. Its distinction is the combination of below-LR prediction, packed source-grid polyphase synthesis, and fused stage-specific precision within this established filtering approach.

Spatial game upscaling. FSR1, NIS, and SGSR1 use local, edge-adaptive reconstruction rather than temporal history [21, 27, 28]. Their integration and reconstruction constraints differ from learned image-restoration models; we include these spatial methods to evaluate that distinct operating regime.

Reparameterization and rendered content. Structural reparameterization adds training capacity while retaining a compact inference graph, as in RepVGG [15]. AEISU folds its kernel-space training coordinates into the deployed operators. QRISP and GameIR provide rendered-content evaluation settings [18, 16]. Unlike common-master resolves, independently rendered LR/HR frames can differ in shading, level of detail, postprocessing, and aliasing; we evaluate these pairing regimes separately.

3 Inverse Supersampling Task

Let $L \in \mathbb{R}^{H\times W\times 3}$ denote the observation and $T \in \mathbb{R}^{2H\times 2W\times 3}$ its reference target. Let $S \in \mathbb{R}^{4H\times 4W\times 3}$ be the common master. For the primary setting, S is 7680×4320, T is 3840×2160, and L is 1920×1080 (width by height).

Define aligned integer-area reduction by s per axis as

$$\mathscr{A}_s(X)[i,j,c] = \frac{1}{s^2}\sum_{u=0}^{s-1}\sum_{v=0}^{s-1} X[si+u, sj+v, c]. \tag{1}$$

In ideal arithmetic the pair is

$$T = \mathscr{A}_2(S), \qquad L = \mathscr{A}_4(S), \qquad \mathscr{A}_2(T) = L. \tag{2}$$

The last identity requires aligned, nested averaging cells and a common numeric domain. Stored images instead have $T = Q_T(\mathscr{A}_2(S))$ and $L = Q_L(\mathscr{A}_4(S))$, where Q_T and Q_L represent quantization and clipping. Independent rounding introduces a measurable baseline discrepancy in $\mathscr{A}_2(T) - L$.

We seek an estimator $\widehat{T} = f_\theta(L)$ fitted to this distribution of pairs. Area averaging removes spatial information, so resolve consistency alone does not identify T. Evaluation therefore measures HR reconstruction rather than LR consistency alone.

Scope and color domain. We distinguish *supersample resolves* of rasterized masters from *integer-area downsampling* of natural images: the averaging operator is the same, but the source distributions differ. Antialiased bicubic reduction defines a different reconstruction problem. Area reduction represents a box-filtered footprint, whereas antialiased bicubic uses a different reconstruction kernel [26, 25]. The intended input is already antialiased, tone-mapped, perceptually encoded RGB in $[0,1]$. AEISU includes no dedicated antialiasing or tone-mapping stage. This placement follows FSR 1's integration contract; GSR 1 also recommends antialiased input for best results [21, 28]. Independently rendered LR frames provide a transfer evaluation outside the sampling model in Eq. (2). Encoded-RGB area averaging is not equivalent to linear-light integration; the transfer function, stored range, and resolve ordering affect the observation model. Section 6.1 describes training acquisition and its limitations.

4 Architecture

4.1 Overview and notation

Figure 1 summarizes the controller and reconstruction path. AEISU maps L to eight analysis channels, packs spatial samples into channels, and predicts reconstruction coefficients with a multiresolution convolutional controller. All its feature-map convolutions operate at $H/2\times W/2$ or $H/4\times W/4$. Analysis arithmetic and reconstruction still operate at the input grid and output phases. Dimensions not divisible by four are handled by padding and cropping.

Let $\mathscr{U}_2$ denote pixel unshuffle (space-to-depth by two), $\mathscr{S}_2$ its pixel-shuffle inverse, and $\sigma(x) = \max(x,0) + 0.1\min(x,0)$. Pixel unshuffle rearranges samples without averaging them. We write $C^k_{a\to b}$ for an affine $k\times k$ convolution with a input and b output channels. The controller uses unit stride, biases, and zero padding for spatial convolutions. The analysis stencils and final reconstruction use replicated source borders. Spatial coordinates and offsets use row–column order, consistent with $\phi = (\phi_y, \phi_x)$.

4.2 Learned analysis inputs

For a source RGB vector $l(p) \in \mathbb{R}^3$, a learned matrix $M \in \mathbb{R}^{3\times 3}$ forms three transformed color channels,

$$z(p) = Ml(p), \qquad y(p) = m^{\mathsf{T}} l(p), \quad m \in \mathbb{R}^3. \tag{3}$$

The scalar y is a learned color mixture initialized with luminance weights. M is initialized to identity, and m is initialized to $(0.2126, 0.7152, 0.0722)$. Four learned, zero-sum directional 3×3 stencils produce responses $d_x, d_y, d_{45}, d_{135}$ from y. For example,

$$d_x(p) = \sum_{j=-1}^{1} \alpha_{j+1}\big[y(p+(j,1)) - y(p+(j,-1))\big]. \tag{4}$$

The remaining operators exchange or combine axis and diagonal differences (Supplementary Material, Section S1). Each starts with a Sobel-like response and learns three parameters controlling directional weighting and strength. The feature pack is

$$E(p) = \big[z_1, z_2, z_3, y, d_x, d_y, d_{45}, d_{135}\big](p). \tag{5}$$

The analysis uses 24 global learned parameters. Native execution fuses analysis with the packed projection; the RGB transform can be absorbed into its weights. Reconstruction still samples the original RGB image L. Section 8.2 evaluates a frozen-analysis control.

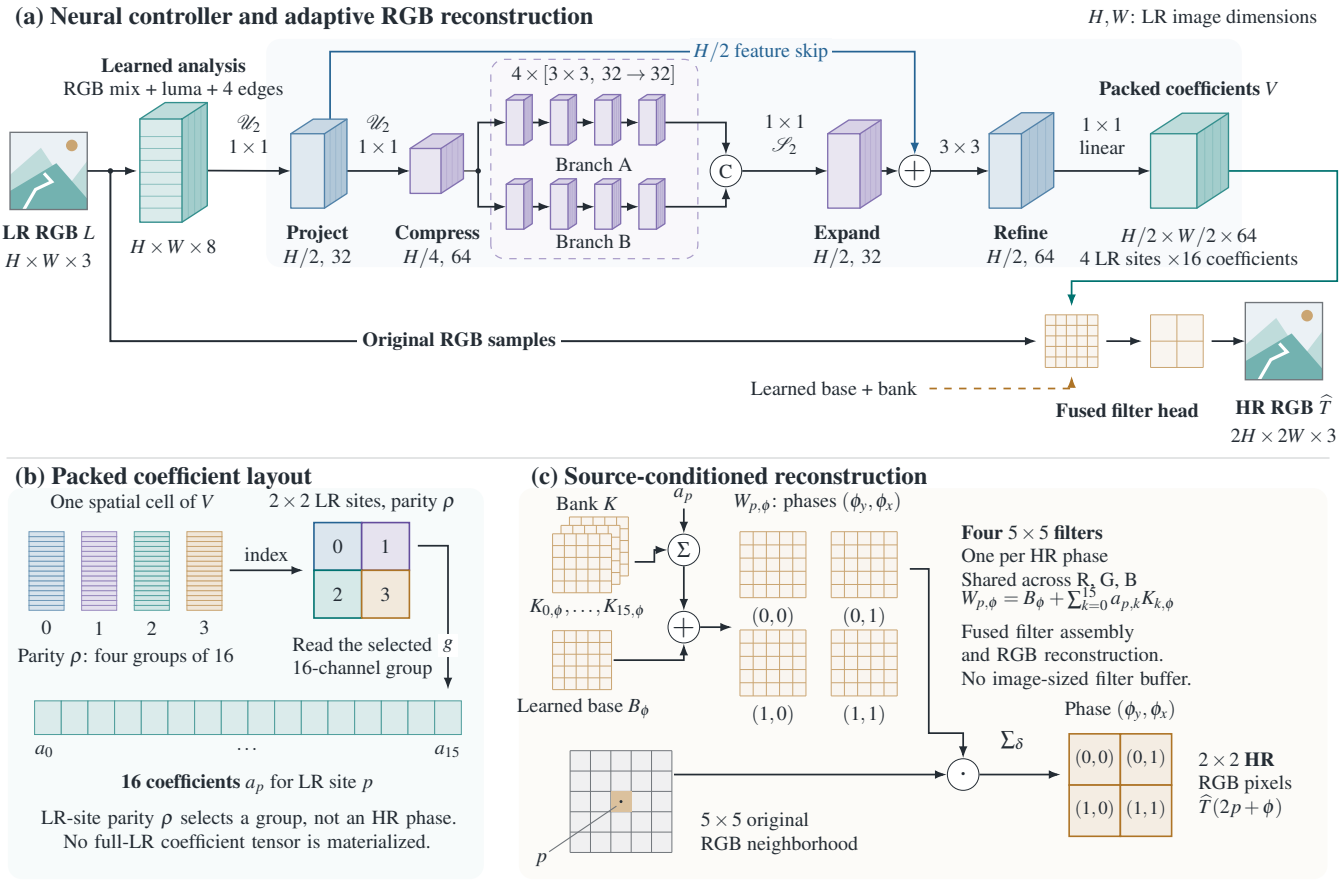


Figure 1: Inference architecture. **(a)** Learned analysis feeds two quarter-resolution branches and a half-resolution skip; original RGB samples bypass the controller. **(b)** The head packs sixteen coefficients for each of four LR-site parities, followed by clamp g. **(c)** Each coefficient vector combines the learned base and bank into four 5×5 phase filters applied to the source RGB neighborhood. Filter assembly and reconstruction are fused. $\mathscr{U}_2$: pixel unshuffle; $\mathscr{S}_2$: pixel shuffle; C: concatenate. Convolutions use LeakyReLU(0.1) except the linear head. Tensor sizes are annotated; icons are schematic.

4.3 Controller below the input resolution

The half-resolution encoding is

$$F_{1/2} = \sigma\big(C^1_{32\to32}(\mathscr{U}_2(E))\big), \tag{6}$$

and the quarter-resolution encoding is

$$Z_{1/4} = \sigma\big(C^1_{128\to64}(\mathscr{U}_2(F_{1/2}))\big). \tag{7}$$

The dense compression learns how information is distributed into the two subsequent 32-channel branches. Let $(Z^a_0, Z^b_0) = \mathrm{split}_{32}(Z_{1/4})$. Each branch has four independently parameterized spatial layers:

$$Z^j_i = \sigma\big(C^{3,j,i}_{32\to32}(Z^j_{i-1})\big), \quad j \in \{a,b\},\ i = 1,\dots,4. \tag{8}$$

There is no cross-branch mixing inside these four layers. Concatenation and a dense expansion mix their final representations:

$$U_{1/2} = \mathscr{S}_2\left(\sigma\left(C^1_{64\to128}\big(\mathrm{cat}(Z^a_4, Z^b_4)\big)\right)\right), \tag{9}$$

$$F = \sigma\big(C^3_{32\to64}(F_{1/2} + U_{1/2})\big). \tag{10}$$

F is the shared feature tensor used for the output-head ablations.

The branch design preserves the spatial-convolution MAC count of a reference core comprising two $64 \to 64$ spatial layers:

$$2 \cdot 4 \cdot 32^2 \cdot 9 = 2 \cdot 64^2 \cdot 9 = 73{,}728 \tag{11}$$

MACs per quarter-resolution site. The split increases depth while retaining two independent feature groups. Section 8.2 compares this deeper, grouped core with the equal-MAC dense alternative.

4.4 Packed coefficient head

A linear 1×1 head maps F to

$$V = C^1_{64\to64}(F) \in \mathbb{R}^{H/2 \times W/2 \times 64}. \tag{12}$$

For an LR site $p = (i, j)$, define $q = (\lfloor i/2 \rfloor, \lfloor j/2 \rfloor)$ and $\rho(p) = 2(i \bmod 2) + (j \bmod 2)$. The raw coefficient vector is indexed directly from the packed head:

$$\tilde{a}_{p,k} = V[q, 16\rho(p) + k], \qquad k = 0,\dots,15. \tag{13}$$

ρ identifies an *input-site parity* within a 2×2 LR block. It is distinct from the four *output phases* generated for each individual LR site. The runtime can use this indexing directly, without materializing a full-resolution coefficient image.

A symmetric soft coefficient clamp maps raw predictions to $a_{p,k} = g(\widetilde{a}_{p,k})$:

$$g(t) = \operatorname{sgn}(t) \begin{cases} |t|, & |t| \le 2, \\ 2 + 0.05(|t| - 2), & 2 < |t| \le 12, \\ 2.5, & |t| > 12. \end{cases} \tag{14}$$

Coefficients pass unchanged within $[-2, 2]$, grow with slope 0.05 beyond this interval, and saturate at ± 2.5. The clamp bounds adaptive modulation; it does not constrain reconstructed RGB to the local source range. Section 8.3 compares this transfer with unrestricted coefficients.

4.5 Polyphase reconstruction

Let $\mathscr{D} = \{-2, -1, 0, 1, 2\}^2$ be the 5×5 LR support, and let $\phi = (\phi_y, \phi_x) \in \{0, 1\}^2$ identify the four HR phases. The model stores a learned base $B_\phi[\delta]$ and sixteen adaptive components $K_{k,\phi}[\delta]$. For each LR site it synthesizes

$$W_{p,\phi}[\delta] = B_\phi[\delta] + \sum_{k=0}^{15} a_{p,k} K_{k,\phi}[\delta]. \tag{15}$$

The output is

$$\widehat{T}[2p + \phi, c] = \sum_{\delta \in \mathscr{D}} W_{p,\phi}[\delta] \, L[\pi(p + \delta), c], \tag{16}$$

where π clamps an out-of-bounds coordinate to the image border. One coefficient vector controls all four output phases. The synthesized spatial weights are shared across the three RGB channels, although the coefficient predictor observes color information. The adaptive bank contains $16 \cdot 4 \cdot 25 = 1{,}600$ values and the base contains $4 \cdot 25 = 100$.

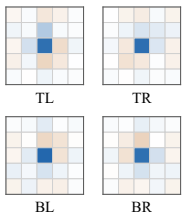
For each component and phase, the effective kernel is obtained from an unconstrained vector $D_{k,\phi}$ by

$$\bar{D}_{k,\phi}[\delta] = D_{k,\phi}[\delta] - \frac{1}{25} \sum_{\delta'} D_{k,\phi}[\delta'], \tag{17}$$

$$K_{k,\phi}[\delta] = \lambda \frac{\bar{D}_{k,\phi}[\delta]}{\max\left(\sum_{\delta'} |\bar{D}_{k,\phi}[\delta']|, \varepsilon\right)}, \tag{18}$$

with $\lambda = 0.5$ and $\varepsilon = 10^{-6}$. Each effective component has zero DC response and ℓ_1 norm at most 0.5 in exact arithmetic; its taps may be signed.

Learned upscale base. The four 5×5 base kernels start as bilinear $2\times$ sampling with half-pixel coordinates and remain trainable. The zero-initialized coefficient head makes the initial output bilinear. Learning adapts these phase-filter constants without changing their support. Its weights enter Eq. (15) directly, so deployment does not evaluate a separate base image.



Learned base phases

Blue: +; orange: −

Zero DC suppresses the adaptive term on constant neighborhoods, while the learned base remains unconstrained. Supplementary Material, Sections S2 and S6, provides a coefficient-perturbation bound and a bank visualization; Section 8 tests quantization robustness.

4.6 Training-time reparameterization

Training adds parameter branches to convolution weights, input analysis, the base, and the adaptive bank, then folds them into the deployed operators. The bank's 64 temporary four-phase kernels and 64×16 mixing matrix assemble the same sixteen effective components before normalization. These training-only coordinates introduce no additional image-sized inference branches and preserve the deployed architecture. Supplementary Material gives the kernel-space folding maps.

5 Tensor Core-Oriented Inference

5.1 Shape and work placement

The convolutional controller uses widths 32, 64, and 128 to support regular matrix tiling on Tensor Cores. The reduction dimensions are C_{in} for 1×1 convolution and $9C_{\text{in}}$ for 3×3 convolution. Actual throughput also depends on datatype, occupancy, kernel implementation, and data movement [17].

Table 1 gives $11{,}520HW$ convolution MACs. For 1920×1080 input this is 23.887872 billion MACs. The production representation stores 115,996 scalars: 113,664 convolution weights, 608 biases, 24 analysis parameters, and 1,700 reconstruction values. Training-only coordinates and quantization metadata are excluded; folding the color transform into the first projection changes the equivalent stored representation.

5.2 Fused reconstruction and mixed precision

Equation (15) permits the learned base and adaptive contribution to be accumulated together. The implementation reuses a local RGB neighborhood across output phases and RGB channels, consumes coefficients from the packed head, and stores shared kernel components in a layout convenient for phase-wise access. The conventional-CUDA reconstruction stage assembles and applies filters without writing an image-sized kernel buffer to global memory.

Numerical formats are selected by stage to balance arithmetic, conversion, and memory costs. The mixed-precision configuration uses FP16 analysis/projection and a fused FP16 refinement/head, with FP8 E4M3 operands in compression, the parallel core, and expansion. Reconstruction remains a separate non-FP8 operator. Precision labels refer to operands and stored intermediates; controller accumulation uses FP32. Section 7 describes calibration.

Table 1: Deployed controller dimensions and convolution MACs per input pixel. Shapes omit the batch dimension. LeakyReLU follows each convolution except the linear coefficient head. Counts cover convolution only.

Stage	Operation	Spatial size	Channels	MACs / LR pixel
Analysis	Learned color mixtures and directional stencils	$H \times W$	$3 \to 8$	—
Encoding	$\mathscr{U}_2$; 1×1 projection	$H/2 \times W/2$	$32 \to 32$	256
Compression	$\mathscr{U}_2$; 1×1 projection	$H/4 \times W/4$	$128 \to 64$	512
Parallel core	Two branches, four 3×3 layers each	$H/4 \times W/4$	$2 \times (32 \to 32)$	4,608
Expansion	1×1 projection; $\mathscr{S}_2$	$H/4 \to H/2, W/4 \to W/2$	$64 \to 128 \to 32$	512
Refinement	Skip addition; 3×3 convolution	$H/2 \times W/2$	$32 \to 64$	4,608
Coefficient head	Linear 1×1 projection	$H/2 \times W/2$	$64 \to 64$	1,024
Controller total	13 deployed convolutions			11,520

6 Experimental Evaluation

6.1 Training data and optimization

Training uses 700 frames selected from 7680×4320 game captures: 635 from *Honkai: Star Rail* and 65 from *Red Dead Redemption 2*. Frames were selected by a pixel-diversity heuristic and area-resolved to 1920×1080 inputs and 3840×2160 targets. The documented Star Rail recording pipeline uses OBS and NVENC AV1, rather than lossless renderer buffers. Capture settings and the incomplete historical acquisition record are described in Supplementary Material, Section S3.

The primary model was trained from fresh initialization for 1,000 epochs (875,000 updates), with batch 32, aligned 512×512 HR crops, 256×256 LR content, and a 16-pixel LR context border. Each epoch comprises 875 random-crop batches (28,000 sampled patches). Augmentation samples the eight D_4 transforms: rotations by multiples of 90° and reflections. RGB MSE is optimized with AdamW ($\beta = (0.9, 0.999)$, $\varepsilon = 10^{-8}$, zero weight decay), gradient-norm clipping at 1, and cosine decay from 2×10^{-3} to 10^{-6} without warmup. Training uses BF16 AMP, channels-last tensors, and seed 1234. The adaptive reconstruction uses a custom backward pass: coefficient gradients are evaluated densely, while gradients for the learned base and bank use spatial subsampling at LR-grid stride 16. Supplementary Material specifies the sampling policy. We use the final epoch, without validation-based checkpoint selection. Section 8 uses separate 100-epoch architecture controls and a 1,000-epoch bicubic-training control.

6.2 Expanded QRISP protocol

We evaluate 600 supplied 960×540/1920×1080 pairs across QRISP's AbandonedSchool, ScifiBaseNightStartStop, ScifiFacility, SeaPort, and SunTempleBush environments [18]. Each contributes 120 pairs: 40 from each of three area-resolve agreement strata. Within an environment, pairs are ranked by encoded-Y MSE between supplied LR and floating-point 2×2 area-reduced HR, after a four-LR-pixel border crop. The lowest quarter, middle half, and highest quarter define the most-, middle-, and least-agreement strata. Uniform rank-quantile sampling within strata and a temporal-spacing guard select frames before inference; 55 source-sequence clusters are represented. The agreement score responds to texture, aliasing, alignment, and rendering differences. Equal stratum representation defines a balanced conditional sample rather than a prevalence-weighted QRISP estimate.

Five input conditions share the unchanged Native HR target and frame identities. SSAA/area uses aligned 2×2 averaging of that target, rounded half-up to uint8. Antialiased bicubic uses Pillow BICUBIC reduction of the same target. FXAA and SMAA process the supplied Native LR frame; Native applies no additional preprocessing. The latter three test independently rendered transfer; the first two are common-reference reconstruction controls. Every method receives identical inputs within each condition, including the same Pillow-bicubic degradation rather than MATLAB resizing. QRISP's Native color modality is non-jittered with zero mipmap bias; antialiasing and frame-blurring postprocessing are disabled [18]. FXAA uses NVIDIA 3.11 PC Quality preset 12, subpixel factor 0.75, thresholds 0.166/0.0833, and encoded-RGB luma [23]. SMAA uses 1x High, luma edges and lookup textures, with linear-light blending [22]. Filtered inputs remain floating point.

Comparators and metrics. All thirteen methods appear in Table 2: interpolation, FSR1 EASU with/without RCAS at 0.2 stops, NIS at sharpness 0, SGSR1, RepNet, RT4KSR, QuickSRNet Small/Medium, SPAN, SwinIR-Light, and deployed mixed AEISU. Spatial/interpolation references use FP32; released neural models use full-frame FP16, without retraining, tiling, or ensembles. QuickSRNet uses the standard AIMET Model Zoo $2\times$ checkpoints. Exact checkpoint identities and execution settings are listed in Supplementary Material.

Primary scores are encoded-Y PSNR/SSIM and RGB LPIPS-AlexNet v0.1 [30], computed from clipped floating-point outputs after a two-HR-pixel border crop. For RGB in $[0, 1]$,

$$Y = \frac{65.481R + 128.553G + 24.966B + 16}{255}. \qquad (19)$$

PSNR uses peak 1; SSIM uses an 11×11 Gaussian window ($\sigma = 1.5$), population moments and the valid interior [24, 29].

LPIPS uses full-resolution RGB mapped to $[-1,1]$, without resizing or patch averaging. RGB PSNR/SSIM and uncropped variants are archived. Per-image scores are averaged within an environment, then the five environments are weighted equally. At this evaluation geometry, resident mixed AEISU inference takes 0.132 ms median (0.152 ms p95), measured in the five-trial resolution sweep (Supplementary Material, Section S5).

A post-hoc paired hierarchical bootstrap resamples environments and source-sequence clusters 50,000 times, preserving pairing. We report nominal 95% intervals and approximate simultaneous 95% standardized-bootstrap bands within separate comparison families. In the results below, inferential comparisons use simultaneous bands that exclude zero; point-estimate comparisons are identified as means. Bands containing zero leave the ordering unresolved, without establishing equivalence. The five-environment sample makes these post-hoc bands exploratory. Supplementary Material, Section S4, details comparison families, scene sensitivity, and numerical reproduction.

6.3 Degradation-dependent reconstruction

Area fidelity is AEISU's strongest result: mixed inference reaches 31.338 dB Y PSNR and 0.91436 SSIM, exceeding every comparator on both metrics. The PSNR gain over the closest external method, QuickSRNet Small, is 1.111 dB. Environment gains range from 0.750 to 1.554 dB; omitting any one environment leaves a 1.000–1.201 dB gain. SwinIR-Light, SPAN, and both QuickSRNet variants nevertheless have lower mean LPIPS. QuickSRNet Small's mean LPIPS advantage is 0.00537, with a nominal interval spanning zero.

Bicubic reverses the fidelity ordering: QuickSRNet Small/Medium, SPAN, and SwinIR outperform AEISU on all three metrics. The adaptive-minus-SwinIR PSNR gap shifts by -3.577 dB from area to bicubic (Fig. 3b). Section 8.4 examines this sensitivity by changing AEISU's training inputs. SPAN and SwinIR-Light obtain these LPIPS scores with pixel-loss training [5, 4, 8]. Mean LPIPS is worse on bicubic than on area for every method; the neural models' relative LPIPS gains arise from smaller deterioration.

FXAA and SMAA inputs give AEISU higher SSIM than every comparator. It also has the highest mean FXAA PSNR and improves against most comparators; the small SMAA PSNR gaps from several spatial methods remain unresolved. SPAN/SwinIR have lower mean LPIPS on both conditions, with greater scene dependence on SMAA. Within AEISU, FXAA raises Native PSNR by 0.822 dB while worsening LPIPS by 0.02912.

6.4 Native transfer and edge reconstruction

On Native, AEISU has the highest mean SSIM and lowest mean LPIPS, but loses PSNR to EASU by 0.644 dB (nominal 95% interval approximately $[-0.785,-0.380]$ dB). It is also 0.619 dB below bilinear; every tested learned model has lower mean Native PSNR than bilinear. The deficit occurs in all five environments and corresponds to approximately 18% higher arithmetic mean Y MSE. Conversely, its LPIPS advantage over EASU is 0.04326, with lower LPIPS on every sampled frame. The closer mean LPIPS lead over SwinIR is 0.01180; SwinIR leads two environment means and EASU+RCAS leads SeaPort.

EASU's mean PSNR is only 0.025 dB above bilinear on Native, yet it improves SSIM by 0.01207 and LPIPS by 0.04862. RCAS improves LPIPS in every condition; on Native it lowers EASU PSNR by 0.419 dB while narrowing the mean adaptive LPIPS lead to 0.00451. Neither this lead nor the lead over SwinIR is resolved by the simultaneous bands. Supplementary Material, Section S4.3 reports environment effects and omission sensitivity. NIS has similar mean distortion scores to EASU on most conditions but worse LPIPS; SGSR1 has lower LPIPS than EASU but often lower PSNR.

Qualitative edge reconstruction. Figure 2 compares selected curved-edge regions from GameIR Town05 using target-derived area inputs. AEISU shows visibly reduced bright edge halos and ringing relative to the displayed neural comparators, with faint residual fringing and sharper contours than EASU. An additional GameIR window-surround crop appears in Supplementary Material. These qualitative examples are separate from the QRISP numerical evaluation.

Target-selected edge profiles. A complementary diagnostic samples 40,517 isolated-edge profiles in 570 of the 600 targets, independently of model outputs. On Area, AEISU has 76–83% lower bright excursion and 77–85% lower profile ringing than the six neural comparators. Its mean transition width is 3.042 HR pixels (target: 2.995), with retained plateau contrast 1.0003 times the target's. Bilinear, EASU, and SGSR1 have lower excursion/ringing values than AEISU; several restoration models also have lower values on Bicubic. Native profiles retain the same ordering against the six neural comparators. Supplementary Material, Section S4.9 details profile eligibility and normalization.

We test Gaussian output smoothing ($\sigma = 0.5, 1, 2$ HR pixels) on AEISU and EASU. On Native, AEISU at $\sigma = 1$ gains 0.646 dB PSNR, while SSIM falls by 0.02058 and LPIPS worsens by 0.14045 (Fig. 3a). Every tested setting worsens aggregate LPIPS on Native/FXAA/SMAA; on Area, the same grid lowers both PSNR and SSIM for both methods. These differing metric responses illustrate the perception–distortion trade-off [31]; Supplementary Material, Section S4.10 gives the complete sweep.

6.5 Capacity and area-resolve agreement

On Bicubic, all three metrics improve along the sequence from QuickSRNet Small to Medium, SPAN, and SwinIR. This ordering changes on Native, and QuickSRNet Medium has lower PSNR than Small on the other four conditions. RT4KSR has lower LPIPS than RepNet on area, bicubic,

Table 2: Expanded QRISP evaluation: 600 source pairs, five environments, unchanged Native HR targets. P: encoded-Y PSNR (dB, higher is better); S: encoded-Y SSIM (higher); L: RGB LPIPS-AlexNet v0.1 (lower). Scores average per-image values within environments, then weight environments equally. Shader/interpolation baselines use FP32; released neural checkpoints use FP16; AEISU uses deployed mixed precision. All conditions use the same two-HR-pixel border crop.

(a) Common-reference reconstruction controls

Method	SSAA / area			Bicubic + AA		
	P	S	L	P	S	L
Bilinear	28.957	0.8602	0.2362	28.839	0.8546	0.2610
Bicubic	29.641	0.8841	0.2115	29.584	0.8799	0.2370
FSR1 EASU	29.690	0.8820	0.1806	29.536	0.8757	0.2117
EASU+RCAS	29.632	0.8892	0.1259	29.637	0.8865	0.1505
NIS (0)	29.621	0.8814	0.2054	29.535	0.8766	0.2333
SGSR1	29.335	0.8802	0.1456	29.351	0.8779	0.1583
RepNet	30.148	0.8954	0.1648	30.592	0.8969	0.1900
RT4KSR	29.982	0.8966	0.1461	30.693	0.9048	0.1619
QuickSRNet S	30.227	0.9046	0.1066	31.527	0.9163	0.1225
QuickSRNet M	30.136	0.9050	0.0985	31.838	0.9204	0.1151
SPAN	29.462	0.9032	0.0837	32.598	0.9290	0.0947
SwinIR-Light	29.575	0.9027	0.0782	32.876	0.9324	0.0870
AEISU mixed	31.338	0.9144	0.1120	31.062	0.9077	0.1509

(b) Independently rendered inputs

Method	FXAA input			SMAA input			Native input		
	P	S	L	P	S	L	P	S	L
Bilinear	27.002	0.8091	0.3006	27.086	0.8224	0.2754	27.292	0.8304	0.2526
Bicubic	27.226	0.8254	0.2722	27.148	0.8367	0.2539	27.088	0.8405	0.2411
FSR1 EASU	27.271	0.8246	0.2511	27.212	0.8364	0.2267	27.317	0.8425	0.2040
EASU+RCAS	27.252	0.8316	0.1930	26.979	0.8375	0.1776	26.898	0.8389	0.1653
NIS (0)	27.236	0.8237	0.2684	27.200	0.8355	0.2482	27.233	0.8406	0.2292
SGSR1	27.067	0.8246	0.1997	26.735	0.8301	0.1916	26.680	0.8316	0.1858
RepNet	27.346	0.8323	0.2325	26.942	0.8407	0.2140	26.289	0.8337	0.2167
RT4KSR	27.251	0.8323	0.2099	26.798	0.8377	0.1959	26.099	0.8280	0.2048
QuickSRNet S	27.268	0.8362	0.1785	26.710	0.8407	0.1680	25.822	0.8294	0.1826
QuickSRNet M	27.226	0.8364	0.1729	26.588	0.8396	0.1619	25.588	0.8260	0.1775
SPAN	27.083	0.8357	0.1594	26.269	0.8362	0.1494	24.032	0.8121	0.1887
SwinIR-Light	27.071	0.8345	0.1567	26.287	0.8343	0.1454	25.058	0.8146	0.1725
AEISU mixed	27.495	0.8410	0.1899	27.139	0.8500	0.1708	26.673	0.8473	0.1607

FXAA, and SMAA, but lower mean PSNR on most conditions. The ordering therefore depends on the input degradation as well as model capacity.

Area-resolve agreement sorting narrows the Native adaptive-minus-EASU PSNR deficit from 0.742 dB in the least-agreement stratum to 0.515 dB in the most-agreement stratum. Mean LR gradient energy is 2.53 times higher in the least-consistent group. AEISU's most-to-least PSNR separation is 5.91 dB on Native but also 5.55 dB on the common-reference area control. Area-resolve agreement therefore captures image difficulty as well as rendering differences. Stratum effects are descriptive because some extremes contain frames from only one sequence. Subtracting each frame's Area adaptive-minus-EASU gap from its Native gap yields -2.291 dB overall (nominal interval $[-2.831, -1.756]$). The adjusted point estimates are nearly unchanged across most-, middle-, and least-agreement strata (-2.274, -2.334, and -2.266 dB).

6.6 Standard bicubic reconstruction

To evaluate the same architecture outside the game-capture setting, we train a separate checkpoint on the 800 official DIV2K training pairs with MATLAB-bicubic $2\times$ inputs [32]. It uses RGB L1 loss for 3,000 epochs (915,000 updates), compared with 875,000 updates for the primary game-trained model. Crop size, D4 augmentation, AdamW and fixed-offset stride-16 reconstruction gradients are unchanged; cosine decay from 2×10^{-3} to 10^{-6} spans the full run. Table 3 reports dataset means over all 219 images in Set5 [33], Set14 [34], B100 [35] and Urban100 [36], using the archived MATLAB-style bicubic pairs. The original game-trained mixed deployment remains the model used in the QRISP and runtime studies.

Both DIV2K precision variants have higher mean PSNR/SSIM and lower LPIPS than RepNet and RT4KSR on every set. Relative to FSRCNN [37], the FP16 row's PSNR is lower by 0.056 and 0.023 dB on Set5 and Set14, but higher by 0.116 and 0.517 dB on B100 and Urban100; SSIM is higher

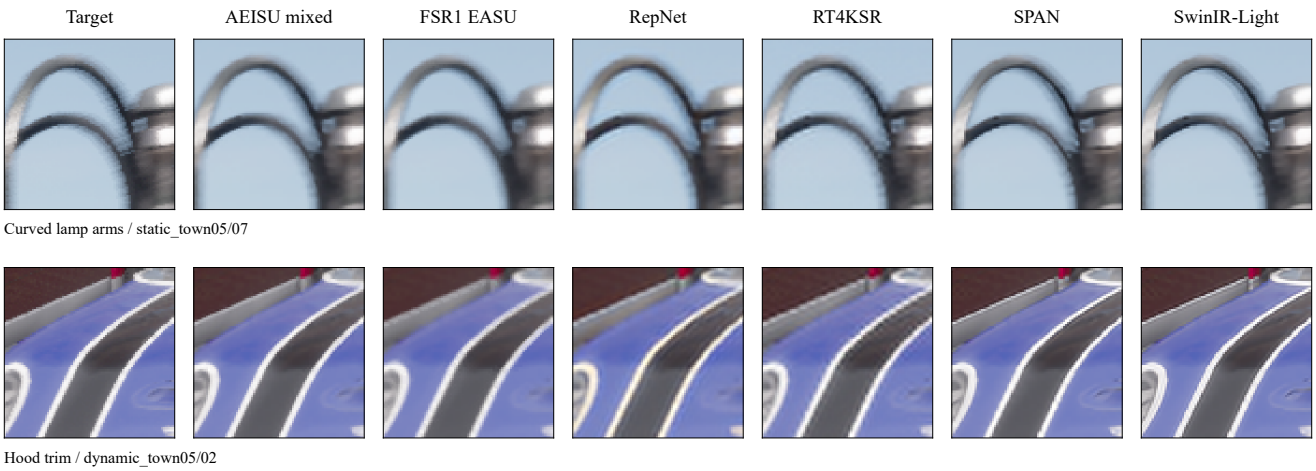


Figure 2: External-method edge reconstruction. Curved lamp arms and hood trim from GameIR Town05, static sequence 07 and dynamic sequence 02, frame 00000030 [16]. All methods use the same target-derived SSAA/area input and unchanged supplied HR reference. Matched 80×80 HR crops retain original colors and use nearest-neighbor enlargement. AEISU uses mixed FP8/FP16, EASU FP32, and neural comparators FP16. GameIR imagery derives from CARLA assets; attribution and license notices accompany the figures.

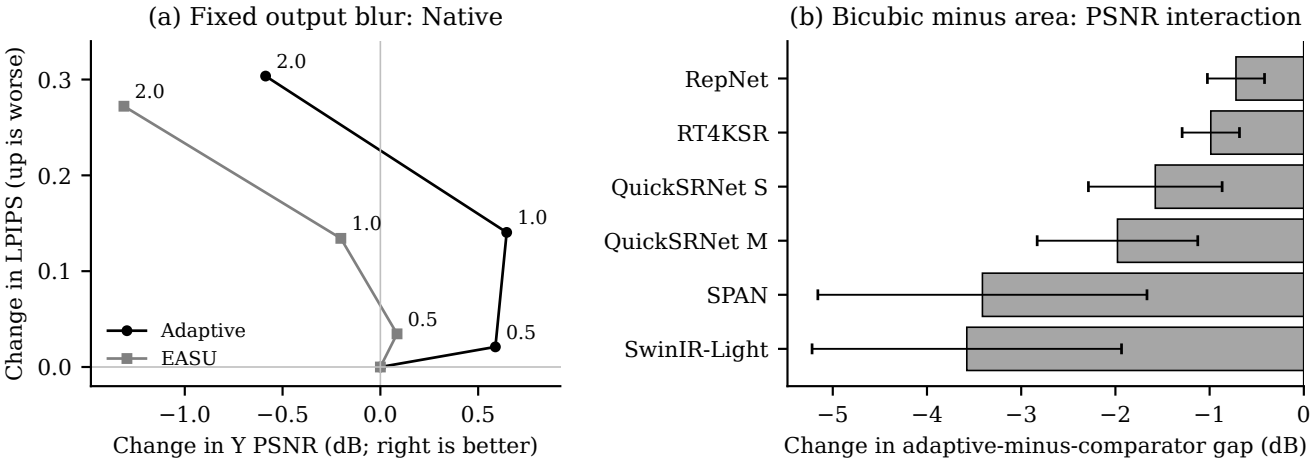


Figure 3: Metric and degradation response. (a) Changes from unblurred Native outputs under the fixed sigma grid; labels denote sigma in HR pixels. PSNR improves to the right, LPIPS worsens upward. (b) Change in the adaptive-minus-comparator Y PSNR gap when replacing area with bicubic inputs on the same references. Negative values favor the comparator's relative response. Bars show separate-family simultaneous 95% bootstrap bands.

on all four, and LPIPS is lower on the latter three. Both variants exceed QuickSRNet Small on all three Urban100 metrics and on B100 PSNR/SSIM, although QuickSRNet Small has lower B100 LPIPS. QuickSRNet Medium, SPAN and SwinIR-Light lead all three metrics on all four sets. Native mixed-FP8/FP16 deployment lowers dataset-mean PSNR by 0.007–0.019 dB and increases LPIPS by 0.00010–0.00095 relative to the common FP16 path. Supplementary Material, Section S4.11, specifies mixed calibration and compares this run with the earlier 1,000-epoch MSE checkpoint.

7 Implementation Measurements

Native measurements use RTX 5090, batch one, CUDA graphs, and five trials of 100 warmup/200 measured iterations. Median and p95 aggregate trial medians and within-trial p95 values. The interval spans analysis through RGB reconstruction, after the input-fence wait and before output signaling; setup, decoding, capture, and presentation are outside it. At 1080p input, halos/alignment pad processing to 1952×1120 before cropping. The harness records hardware state without changing clock or power settings.

Mixed precision reduces median latency by 14.8% relative to uniform FP8 at 1080p. This gain combines datatype, fusion, conversion, and kernel-layout choices. Uniform FP8 scales nonlinearly from 1080p to 1440p: $2.32\times$ latency for $1.78\times$ nominal input pixels. Repeated stage measurements localize much of the excess to analysis/projection and the coefficient head (Supplementary Material, Section S5.2). The FP16

Table 3: Classic bicubic 2× reconstruction. Dataset means of encoded-Y PSNR (P, dB), Y SSIM (S), and RGB LPIPS (L), after a two-HR-pixel border crop. All metrics are recomputed from RGB outputs. Both AEISU rows use the DIV2K L1 checkpoint trained for 3,000 epochs (915,000 updates): FP16 is the common FP16-controller/FP32-reconstruction path; mixed is the native FP8/FP16 deployment. Released comparators retain their recorded precision and training.

	Set5 (5)			Set14 (14)			B100 (100)			Urban100 (100)		
Method	P	S	L	P	S	L	P	S	L	P	S	L
Bilinear	32.258	.91207	.17028	29.253	.84084	.23532	28.688	.81075	.30459	25.969	.80917	.24933
FSR1 EASU	33.865	.93130	.12233	30.457	.86763	.18344	29.572	.83950	.24733	27.208	.84594	.18840
FSRCNN	37.154	.95691	.05846	32.744	.90970	.10668	31.514	.89039	.16085	29.848	.90174	.10580
RepNet	35.848	.94509	.10667	32.003	.89427	.16634	30.907	.87172	.22500	28.800	.87967	.15645
RT4KSR	36.147	.95192	.06915	32.090	.90384	.12241	31.070	.88553	.18500	28.743	.88570	.13466
QuickSRNet S	37.195	.95756	.05686	32.834	.91064	.09947	31.616	.89244	.15133	30.179	.90671	.09428
QuickSRNet M	37.434	.95866	.05718	33.052	.91297	.09875	31.818	.89545	.14914	30.773	.91428	.08246
SPAN	38.099	.96102	.05556	33.706	.91914	.09537	32.235	.90058	.14743	32.212	.92924	.06570
SwinIR-Light	38.183	.96140	.05400	33.870	.92079	.08824	32.325	.90159	.14047	32.785	.93436	.05705
AEISU FP16	37.098	.95741	.06068	32.721	.91098	.10409	31.630	.89269	.15294	30.365	.90969	.08924
AEISU mixed	37.090	.95737	.06078	32.709	.91089	.10444	31.621	.89258	.15389	30.346	.90949	.08975

Table 4: Resident GPU inference time (ms) on RTX 5090. Median and p95 are aggregated over five trials.

Input	FP16	FP8	Mixed	Mix. p95
540p	0.285	0.204	0.132	0.152
720p	0.473	0.308	0.206	0.211
1080p	0.974	0.488	0.416	0.440
1440p	1.677	1.132	0.740	0.764

core uses Tensor Core WMMA; reconstruction uses conventional CUDA arithmetic. The mixed runtime requires Blackwell SM120a and has been measured on RTX 5090. Supplementary Material, Section S5 gives runtime versions, instrumentation, and both timing geometries; expanded quality uses 540p-to-1080p.

Computational and implementation context. Table 5 reports deployed parameters, counted MACs, and common FP16 PyTorch timings at the QRISP geometry. The controller's 11,520 MACs/LR pixel exclude reconstruction: direct filter assembly adds 1,600 and RGB application adds 300, yielding 13,420 before analysis arithmetic and padding. The common protocol uses batch-one full-frame channels-last tensors, cuDNN autotuning, disabled TF32, and five trials of 20 warmup/50 measured forwards, with eager and graph execution separated. Under graph execution, QuickSRNet S takes 0.292 ms and AEISU 1.588 ms, whereas the optimized mixed deployment takes 0.132 ms. RepNet has fewer MACs and parameters, and its 0.129 ms graph latency is close to the optimized AEISU latency. Ordinary PyTorch synthesis materializes per-pixel filters and RGB patches; fused deployment avoids these image-sized intermediates. Figure 4 pairs current quality scores with fresh, backend-labeled timings.

Calibration uses 24 training and six separate calibration-validation images; details are in Supplementary Material, Section S5.1. On 219 area-derived natural images, mixed-to-FP32 output agreement averages 54.594 dB RGB PSNR (minimum 31.658 dB); target PSNR changes by -0.0125 dB on average. The worst target RGB PSNR loss is 0.479 dB; three images lose more than 0.1 dB (Supplementary Material, Section S5.1).

Table 5: Deployed complexity and FP16 PyTorch timing (ms), 540p-to-1080p. MACs count attention, required padding and AEISU reconstruction over the full padded PyTorch canvas; analysis and non-MAC operations are excluded. E: eager; G: graph replay.

Method	Param. (k)	GMACs	E	G
AEISU	116.00	7.67	1.721	1.588
QuickSRNet S	22.86	11.79	0.326	0.292
QuickSRNet M	50.60	26.13	0.611	0.554
RepNet	26.61	3.43	0.173	0.129
RT4KSR	44.52	6.05	1.108	0.509
SPAN	410.70	212.38	5.922	5.820
SwinIR-Light	910.15	546.06	124.587	123.529

Conventional-CUDA deployment. A secondary deployment preserves the folded architecture and weights, using FP16 projection/refinement/head and INT8 compression, parallel-core, and expansion convolutions on conventional CUDA cores. In a matched five-trial comparison, resident 1080p-to-4K latency is 1.275 ms, versus 0.418 ms for the native FP8/FP16 Tensor Core deployment, a 3.05× latency ratio. The latter session's 0.442 ms p95 agrees closely with the repeated resolution sweep's 0.440 ms. Mean target Y PSNR decreases by 0.009 dB on a 60-pair QRISP subset and 0.070 dB on twelve internal 4K validation pairs. This deployment comparison combines instruction path, precision, and fusion choices; detailed timing and quality results are in Supplementary Material.

8 Ablation Studies

We examine output representation, learned analysis/base constants, parallel-core structure, coefficient clamping, and training degradation. The architecture controls use separate 100-epoch runs; the degradation comparison uses two 1,000-epoch checkpoints, with one training run per configuration.

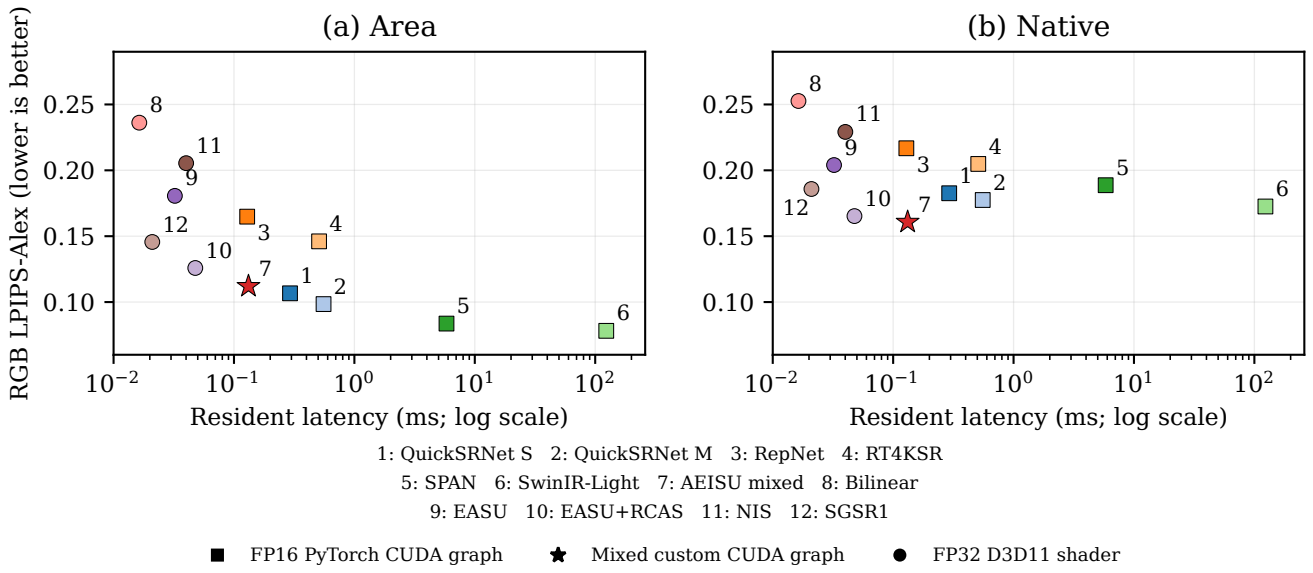


Figure 4: Quality and implementation latency at 540p-to-1080p. Current 600-pair Area and Native RGB LPIPS means are paired with fresh resident RTX 5090 timings. Symbols identify FP16 PyTorch graphs, the custom mixed CUDA graph, and FP32 D3D11 shaders. All timings use five trials; released-neural runs use 20 warmup/50 measured forwards and native/spatial runs use 100/200. Checkpoints, architectures, and shader identities were verified against the quality suite. Different numerical precision and backend optimizations prevent architecture-only speed conclusions.

Each study evaluates final checkpoints with a common implementation within that study. Training differences and their implications are summarized in Section 9 and specified in the supplement.

8.1 Output representation

The backbone was developed under a strict below-input-resolution processing constraint before adaptive reconstruction was adopted. We replace the head after the activated refinement tensor F in Eq. (10), preserving the upstream architecture and training each variant end-to-end. Let $\mathscr{L}_2(L;B_{\text{res}})$ denote reconstruction by the residual model's learned four-phase 5×5 base, initialized to bilinear sampling. Two minimal RGB alternatives are

$$\widehat{T}_{\text{res}} = \mathscr{L}_2(L;B_{\text{res}}) + \mathscr{S}_4\big(C^1_{64\to48}(F)\big), \tag{20}$$

$$\widehat{T}_{\text{direct}} = \mathscr{S}_4\big(C^1_{64\to48}(F)\big). \tag{21}$$

Because F is on the half-LR grid, the head must increase each spatial dimension by four to reach $2H\times2W$; hence $3\cdot4^2=48$ output channels.

The three tested heads are learned-base adaptive filtering, RGB residuals on a learned base, and direct depth-to-space RGB reconstruction without a base. Each model was trained on the same 700 pairs for 100 epochs and 87,500 optimizer updates, with batch 32, HR patches of size 512, and cosine decay from 2×10^{-3} to 10^{-6}. The coefficient and residual projections start at zero, giving their bilinear-initialized bases; the direct head starts at zero RGB. The adaptive bank retains its sparse gradient estimator, while the direct heads use dense autograd. The coefficient clamp in Eq. (14) applies to the adaptive head.

Precision protocol. The FP8 stress test uses E4M3 weights and activations in all nine sequential convolutional stages; each of the four core stages contains two parallel convolutions, giving thirteen convolutions in total. Weights have per-output-channel scales; activations have per-tensor scales. Accumulation and learned reconstruction constants remain FP32, and native image I/O and linear-head outputs remain FP16. All models use the same calibration images, procedure, and budget. The RGB projections are zero-padded from 48 to 64 channels for native execution.

Quantization robustness. Adaptive filtering achieves higher FP16 reconstruction quality and smaller PSNR degradation under FP8 quantization on both datasets (Table 6). On QRISP SSAA, its FP16 Y PSNR exceeds residuals by 0.273 dB and depth-to-space by 0.755 dB. FP8 reduces Y PSNR by 0.051 dB, versus 0.216 and 1.107 dB, respectively. Agreement between each model's FP8 and FP16 outputs follows the same ordering, indicating smaller numerical perturbations for adaptive filtering. Supplementary Material, Section S2 gives a coefficient-level mechanism consistent with the observation.

Haloing and edge reconstruction. Figure 5 shows narrower fringes for adaptive reconstruction in four selected validation regions, with the widest fringes in the depth-to-space output. Faint fringing remains in the adaptive output. The

Table 6: Output-head quality and uniform-FP8 sensitivity: 60 QRISP AbandonedSchool Enhanced targets and 219 natural images, with area-derived inputs and uncropped Y metrics. Loss: FP16 minus FP8 target PSNR. Agreement: RGB PSNR between each model's FP8 and FP16 outputs.

Track	Head	Y PSNR (dB)		Y SSIM		Loss (dB)	Agreement (dB)
		FP16	FP8	FP16	FP8		
SSAA game	Adaptive	32.874	32.823	0.90149	0.90083	0.051	49.81
	RGB residual	32.601	32.385	0.89709	0.89270	0.216	46.30
	Depth-to-space	32.119	31.012	0.88841	0.85247	1.107	35.84
Natural images	Adaptive	30.690	30.624	0.89737	0.89630	0.066	46.55
	RGB residual	30.327	30.160	0.89149	0.88503	0.167	43.83
	Depth-to-space	29.535	28.312	0.87818	0.82152	1.223	32.65

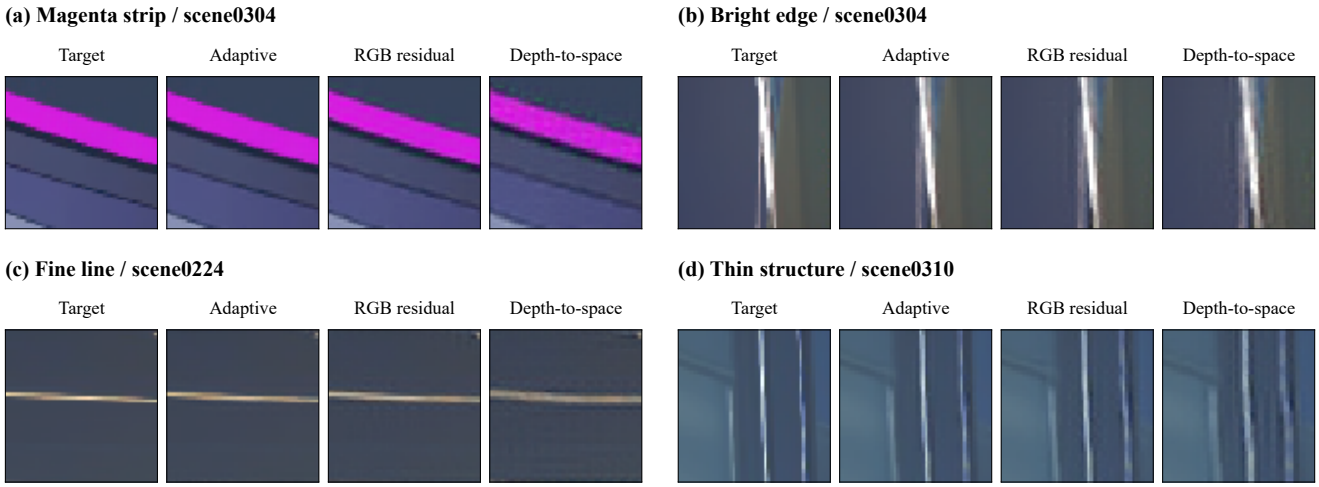


Figure 5: Output-head edge reconstruction. UI-free *Honkai: Star Rail* validation crops: (a) magenta strip, (b) bright edge, (c) fine line, (d) thin structure. Columns: target, adaptive filtering, RGB residuals, depth-to-space. Nearest-neighbor enlargement; original colors.

panels use FP16-AMP tiled outputs with matched crops and original colors.

Effect of the output representation. For 1080p input, the final feature tensor is at 540p. Direct RGB heads must reconstruct fine structure from these compressed features. The adaptive head instead conditions filters applied to original LR neighborhoods; the residual control samples LR through its base, but predicts the correction from half-LR features. This distinction offers a plausible explanation for the observed fidelity and haloing differences on the shared backbone. The tested adaptive configuration improves reconstruction without adding full-LR convolutional feature stages.

8.2 Core architecture controls

Three additional checkpoints freeze the analysis mixtures and stencils, freeze the bilinear-initialized base, or replace the parallel core with two dense 64-channel convolutions. Each uses the head study's 100-epoch recipe and seed 1234; the dense alternative matches Eq. (11)'s bottleneck MACs. All four use common FP16-controller/FP32-reconstruction evaluation on the frozen 600-pair sample.

Table 7: 100-epoch core controls on Area. Training MSE is scaled by 10^4; Y metrics and RGB LPIPS use 600 QRISP pairs and a two-HR-pixel crop. Lower LPIPS is better. The reference differs from the longer-trained mixed deployment.

Control	Train MSE	Y PSNR	Y SSIM	LPIPS
Reference	1.1471	31.297	0.91375	0.11390
Fixed inputs	1.1506	31.304	0.91395	0.11219
Fixed bilinear base	1.1566	31.283	0.91346	0.11395
Dense 64×2 core	1.1578	31.277	0.91370	0.11396

Learning the analysis and base slightly lowers final training MSE, while external FP16 differences are small and metric-dependent (Table 7). Fixed inputs instead improve all three Bicubic metrics. The parallel-core checkpoint has higher Area PSNR than the dense control by 0.020 dB. Under controller E4M3 rounding, the reference loses 0.038 dB PSNR, versus 0.053 for fixed inputs, 0.077 for the fixed base and 0.075 for the dense core. Relative to the reference's precision change, the fixed base has greater LPIPS degradation and the dense core has greater SSIM and LPIPS degradation. Supplementary Material, Section S4, gives all-condition, RGB, and precision results.

8.3 Coefficient clamp

An additional checkpoint replaces g in Eq. (14) with the identity, retaining the 100-epoch recipe and evaluating both checkpoints through common FP16-controller/FP32 tiled reconstruction on all 600 pairs. Training MSE rises from 1.1471×10^{-4} to 1.1664×10^{-4}. Without the clamp, Area Y PSNR falls from 31.297 to 31.253 dB and Bicubic PSNR falls by 0.028 dB; LPIPS also worsens on Bicubic and FXAA. The exception is a small 0.006 dB SMAA PSNR gain. Under the core controls' uniform-controller E4M3 rounding test, Area PSNR loss increases from 0.038 to 0.069 dB, with an additional 0.032 dB loss and greater LPIPS degradation. Both typical and extreme raw coefficient magnitudes are larger without the clamp (Supplementary Material, Section S4).

8.4 Training degradation

With the same 1,000-epoch budget and targets, bicubic retraining raises Bicubic Y PSNR by 0.665 dB but lowers Area and Native PSNR by 0.740 and 0.718 dB in common FP16 evaluation. LPIPS improves on Area, Bicubic and FXAA. The Bicubic run uses direct 8K-to-1080p resizing, whereas benchmark bicubic inputs are derived from HR targets (Supplementary Material, Section S4). On Bicubic, the retrained checkpoint reaches 31.725 dB, still below SPAN (32.598) and SwinIR-Light (32.876), and also has higher LPIPS.

9 Limitations and Discussion

AEISU is single-frame and fixed at $2\times$, with 5×5 direct RGB reconstruction support. Signed filters and an unconstrained base permit overshoot. The profile diagnostic covers eligible isolated edges with approximately flat flanks; Native profiles also reflect differences between independently rendered LR/HR images. Temporal stability and whole-image halo prevalence remain unevaluated.

All training controls use one seed. The head study shares the backbone but varies initialization, gradient estimation, and constraints, and the degradation study uses different master/target resizing chains. These differences limit attribution of the measured effects to individual design choices. Repeated seeds and a full-LR feature-processing control would strengthen that attribution.

Primary training uses captures from two games and final-checkpoint selection without internal validation; the separate natural-image checkpoint uses DIV2K. Incomplete scene-level acquisition records leave the overlap of existing same-game 4K diagnostic frames with training scenes unknown. QRISP covers five environments with equally weighted agreement strata; released comparators retain their own training data and objectives. The main quality study is 540p-to-1080p, while 1080p-to-4K evidence consists of timing and internal diagnostics.

The smoothing and agreement controls leave the contributions of aliasing, misalignment, rendering differences, and degradation specialization unresolved. Runtime results apply to the evaluated implementations on one GPU and measure resident inference rather than application latency.

10 Conclusion

We presented AEISU, a compact $2\times$ upscaler that predicts local filters below the input grid and applies them to original RGB samples. The optimized mixed deployment reaches 0.416 ms median for resident 1080p-to-4K inference on RTX 5090. Shared-backbone controls favor the tested adaptive head for reconstruction fidelity and FP8 robustness. Area-resolved inputs are the model's strongest setting; its fidelity and LPIPS rankings change under bicubic and independently rendered inputs.

Supplementary material. The supplement and companion repository provide training and folding details, evaluation protocols, measurements, and visualizations; image corpora are excluded.

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